

The effect of strong magnetic fields on small-scale dynamics

Candidate # 3

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Abstract

We investigate the influence of strong magnetic fields on small-scale convective dynamics in and around a sunspot located in active region AR12770. The observations were obtained on August 7, 2020, between 08:22:21 - 08:59:58 UT, with the Swedish 1-m Solar Telescope. Using spectroscopic measurements of the Fe I 617.3 nm line, we analyze Doppler velocities and spectral line characteristics across three subfields: quiet Sun, penumbra, and umbra. To characterize the spectral lines, both Gaussian and Voigt profiles were fitted to the data. We find that Gaussian profiles offer an efficient and sufficiently accurate means for measuring Doppler shifts, particularly in regions where the line core dominates. However, Voigt profiles are more suitable when detailed modeling of the line wings is necessary, especially in areas affected by significant magnetic broadening. Our findings indicate a suppression of convective motion in regions of stronger magnetic fields, accompanied by broader and deeper absorption lines, likely due to a combination of temperature effects and Zeeman broadening. The equivalent width in the quiet Sun measured $W_\lambda = 68.538 \text{ mÅ}$, but in the umbra the equivalent width was increased to $W_\lambda = 114.52 \text{ mÅ}$. Doppler maps reveal strong upflows and downflows in the quiet Sun that are absent or diminished in magnetized regions. While the data support magnetically induced modifications to plasma dynamics and line formation, further analysis is required to disentangle the contributions from temperature, viewing angle and magnetic field strength. We cannot draw a final conclusion on the correlation between equivalent width and Doppler velocities, but we draw attention to an interesting trend observed, which requires additional studies.

Keywords: Sun: sunspot, magnetic fields, sunspot.

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1. Introduction

In late-type stars like the Sun, surface convection plays a fundamental role in transporting energy from the stellar interior to the outer layers. The surface convection is responsible for the granulation pattern directly observed in high-resolution pictures of the Sun (Al-lende Prieto et al. 2013). The presence of magnetic fields, particularly strong ones, can significantly alter the nature of convective motion at the solar surface. This effect is especially prominent in and around sunspots—regions of intense magnetic activity, where both the intensity and dynamics of convection are visibly altered. In this study, we investigate how the magnetic fields associated with a sunspot and its surrounding regions influence convective motions observable in the solar photosphere.

Convection is one of the main transport mechanisms in the Sun. It occurs when the temperature gradient is too large, making the region unstable to convection. In the early days, turbulent convection was described in terms of the mixing-length theory (MLT, Böhm-Vitense 1958), where the mixing length is the assumed distance a parcel of gas can move up or down before it dissolves into the new environment, thus releasing excess heat. The mixing length is considered to be proportional to the pressure scale height. Note that MLT is a simple model of turbulent convection. It is a local theory and does not allow for overshoot of convective motions into surrounding layers. This limits the ability of MLT to fully describe convective processes.

In recent years, scientists have utilized numerical approaches to model and deepen their understanding of convection. Two philosophies have emerged, where one favors simplified physics (e.g. [Hurlburt & Toomre 1988](#), [Weiss et al. 1996](#), [Brun et al. 2004](#)) to fully explore the roles of different effects and parameter regimes. The other attempt to implement as realistic physics (e.g. [Nordlund 1985](#), [Carlsson et al. 2004](#), [Stein & Nordlund 2006](#)) as possible at a higher computational expense in favor of more accuracy with observations ([Priest 2014](#)).

Convective motions are seen as cellular features covering the photosphere, the Sun's thin visible surface layer. This pattern is called granulation and consists of irregular-shaped bright cells called granules surrounded by dark intergranular lanes. The center of each granule is bright due to rising, hot plasma. At a certain position, the opacity is not sufficient to prevent the escape of radiation. Heat is effectively removed by radiation, causing the plasma to become cool and advected sideways before descending the intergranular lanes. Granules typically have a diameter of 1000 km, 1 km/s^{-1} vertical flow and 0.2 hr turnover-time ([Rast 2003](#)).

We find a large concentration of magnetic activity in the solar active regions in the form of sunspots. They are structures characterized by a dark central region, the umbra, surrounded by a brighter filamentary penumbra. [Hale \(1908\)](#) first measured a magnetic field in sunspots, leading to a breakthrough in understanding the nature of sunspots. This established magnetic field as the cause of the sunspot phenomenon ([Nordlund et al. 2009](#)). The umbra has an almost vertical magnetic field, and the filamentary structure has a weaker, more horizontal magnetic field. They evolve continuously and the number of sunspots is known to be connected to the solar cycle of approximately 11 years. [Borrero et al. \(2017\)](#) found the average quiet Sun is about $(374 \pm 95) \text{ K}$ hotter than the average penumbra at $\tau_5 = 1$. When comparing only to the bright penumbral filaments, the average quiet Sun is $(257 \pm 95) \text{ K}$ hotter than the average over the bright filaments. The umbra, which is the coolest structure observed in the solar photosphere, is around 200 K lower than the quiet Sun photosphere ([Kiess et al. 2014](#)). The width of penumbral filaments measures 0.37 arcsec ([Bonet et al. 1982](#)) and mean length reaches about 5 arcsec ([Tiwari et al. 2013](#)). This translates to a width of around 268.2 km and an extension of 3625 km, compared to a granule with a mean diameter of 1050 km ([Abramenko et al. 2012](#)).

A sunspot appears dark or rather less bright than its surroundings due to partial inhibition of convective energy transport due to its strong magnetic field ([Priest 2014](#)). [Hoyle \(1949\)](#) pointed out that it is mainly motions across the field lines that are suppressed. The suppression of convective motion is expected to alter the overall temperature stratification in the overshoot region ([Fan 2021](#)). The magnetic field strength also alters the plasma flow and how the convection manifests itself on the photosphere. [Löhner-Böttcher et al. \(2018\)](#) found that the convective blueshift in a sunspot umbra is significantly reduced, measuring around -30 m/s , compared to approximately -350 m/s in the quiet Sun. [Aparna et al. \(2025\)](#) found a decrease in horizontal advection speed of magnetic flux with an increase in B_z . From $110 \pm 3 \text{ m/s}$ in the network and plage region to $10 \pm 4 \text{ m/s}$ for 2500 G in the sunspot umbra. This highlights the effect of a strong magnetic field being present. As an explanation for the brightness in the penumbra, [Spruit & Scharmer \(2006\)](#) suggested that field-free gaps in the inclined magnetic field of the penumbra may be the reason for elongated bright filaments rather than point-like dots. The surplus brightness of the penumbra relative to the umbra might be since convective cells can become larger in the more inclined and weaker magnetic field, as in the more vertical and stronger field of the umbra ([Rempel & Schlichenmaier 2011](#)).

The convection in the penumbra is not entirely suppressed, though the magnetic field alters its nature, leading to filamentary structures and flows. The horizontal flows is seen as long strands called filaments, both bright and dark. In the dark filaments, the magnetic field is almost horizontal at the other edge of the penumbra. In the bright filaments, the inflation is approximately 35° , making the mean field included by around $15\text{-}20^\circ$ ([Tildesley 2003](#)). We analyse convective motion in and near a sunspot based on high-resolution observations from the Swedish 1-m Solar Telescope (SST). The aim is to study how strong magnetic fields affect small-scale dynamics. This work enhances our understanding of magneto-convection by quantifying how magnetic fields suppress or modify convective flows. The results contribute to improving physical models of sunspots and active regions, aiding in the interpretation of solar spectropolarimetric data.

In this work, we first describe the observational dataset and methodology used to probe the effects of magnetic fields on convective motions in [section 2](#). This includes details on the instrumentation, data reduction and the analysis techniques applied to extract Doppler velocities and spectral line properties. In [section 3](#), we present the results from the three subfields with different magnetic field strengths, highlighting differences in intensity, Doppler velocity and spectral characteristics. We further interpret these findings in [section 4](#), where we compare our results with previous studies and consider the implications of magnetic suppression of convection. Finally, we summarize our conclusions and suggest directions for future work in [section 5](#).

83 2. Data and methods

84 2.1. Instrumentation

85 The observations were obtained with the CRisp Imaging Spectropolarimeter (CRISP, [Scharmer et al. 2008](#)) at the Swedish 1-m
86 Solar Telescope (SST, [Scharmer et al. 2003a](#)) on La Palma. CRISP and the SST together yield imaging spectropolarimetry at unsur-
87 passed high spatial, spectral and temporal resolution ([Vissers et al. 2013](#)). CRISP employs a dual Fabry-Pérot interferometer, which
88 enables wavelength tuning within 50 ms, operating within the red part of the spectrum. Light is first guided through an optical
89 chopper, which ensures strict synchronization between the cameras. The three CCD cameras are identical high-speed, low-noise
90 Sarnov CAM1M100 cameras with 1 K x 1 K chips. A prefilter wheel provides various wavelength bands, allowing for a change
91 within 250-600 ms. Between the prefilter wheel and CRISP, some of the light is branched off to a camera imaging this wide band to
92 serve as "multi-object" anchor in the post-processing.

93 To counteract atmospheric seeing, the SST is equipped with excellent adaptive optics (AO, [Scharmer et al. 2003b](#)). Seeing is a
94 disturbance of light due to the turbulent atmosphere of the Earth. The AO utilizes a wave-front sensor that measures the seeing
95 conditions across the field of view (FOV). A moving flat tip-tilt mirror corrects for the large-scale distortion in real-time. Additional
96 corrections for small-scale distortions are achieved by a deformable mirror with the capability to change its shape at specific location.

97

98 2.2. Reduction

99 The SST data were reduced using the SSTRED reduction pipeline ([Löfdahl et al. 2021](#)), which is a further development of the
100 CRISPRED pipeline ([de la Cruz Rodríguez et al. 2015](#)). The application of image restoration with the multi-object multi-frame
101 blind deconvolution (MOMFBD, [Van Noort et al. 2005](#)) is an important step of the SSTRED pipeline to produce high-quality
102 science-ready data. MOMFBD reduces the remaining high-order image deterioration from atmospheric seeing considerably ([Vis-
103 sers et al. 2013](#)), after the flat tip-tilt mirror and deformable mirror. It operates under the assumption that the recorded image consists
104 of the true image convolved with a space-invariant point spread function (PSF) and corrupted by random noise.

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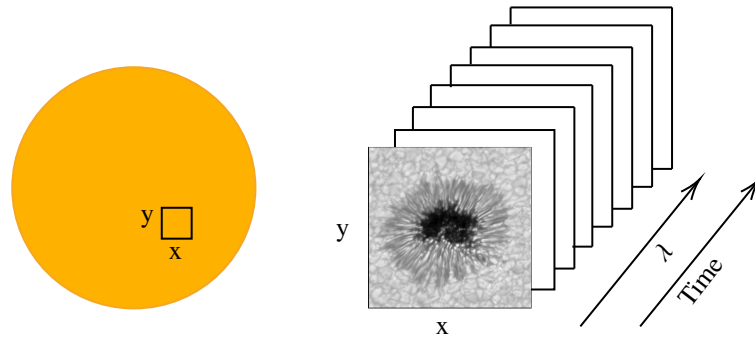


Fig. 1. Schematic illustration of the data set obtained from SST. Each 2D slice (seen on the right) captures spatial information on the solar surface (seen on the left). The remaining dimensions contain temporal and wavelength information, allowing for the study of dynamic evolution.

106 2.3. Observational data

107 A Fe I 617.3 nm sequence was acquired on 2020-Aug-07 between 08:22:21 and 08:59:58 UT, covering active region AR12770
108 containing a sunspot. The FOV was located at viewing angle $\mu = 0.83$, centered on $(X, Y) = (-446, 279)$ in standard heliocentric
109 solar coordinates (in arcseconds, with positive X -direction pointing west and positive Y -direction pointing north of the apparent
110 solar disk). The Fe I line was sampled at 14 wavelength points from 617.301 nm to 617.401 nm. The temporal cadence was about
111 40 s and the full FOV covers 512×512 pixels ($29.''638 \times 29.''638$).

112 Within the full FOV, we chose three subfields to use in a comparative study. The three subfields are located in regions with varying
113 magnetic field strengths, and we aim to quantify the differences in LOS velocity and spectral line shape as a result of these magnetic
114 field strengths. We use the diffraction limit as a reference to determine the size of the subfields. The diffraction limit sets a theoretical
115 limit for the achievable angular resolution of the telescope, which is the ability of the telescope to spatially resolve two closely spaced
116 objects and is dictated by the propagation of the light wave. For a given wavelength λ which a telescope with diameter D observes

at, its diffraction limited resolution θ is given by the Rayleigh criterion,

$$\Delta\theta'' = \left(1.22 \frac{\lambda}{D}\right) \cdot (60)^2 \cdot 180/\pi. \quad (1)$$

However, this theoretical limit is rarely achieved by ground-based telescopes. The actual angular resolution is determined by the optical aberrations and seeing conditions (Cicone 2022). For our observations $\theta \approx 0.1602''$, meaning the telescope can theoretically resolve features of size $0.1602''$. Thus, we ensured the subfields are larger than the theoretical angular resolution. Overview of the

Subfield	Location	Coordinates
A	Quiet Sun	(-435.5, 289.5)
B	Penumbra	(-451.5, 273.5)
C	Umbra	(-445.6, 279.4)

Table 1. Overview of the three chosen subfields on the Sun used in this comparative study, along with their location and coordinates. The given coordinates are where the FOV for each subfield is centered, in standard heliocentric coordinates.

chosen subfields can be found in Table 1. Subfield A is located in the quiet Sun with a visible granulation pattern, subfield B is located in the sunspot penumbra and subfield C, located in the sunspot umbra. All subfields cover 86×86 pixels corresponding to a physical extent of $4.''988 \times 4.''988$. For visual context, see Figure 2 for an intensity map at the initial time step. In the figure, we see the full FOV containing the sunspot in the upper left corner with red boxes marking the location of each subfield. Subfields A, B and C are shown in the upper right corner, lower left corner and lower right corner, respectively.

2.4. Doppler map

To investigate the convective motions of the plasma, more specifically the upflows and downflows, we analyzed Doppler velocities within the subfields. The aim was to study the temporal evolution of the plasma movement and how it differs between the subfields due to magnetic fields. We calculated the Doppler shift for all considered pixels and time steps using Equation 2,

$$v = \frac{\Delta\lambda}{\lambda_0} \cdot c. \quad (2)$$

λ_0 represents the central wavelength for the spatially averaged intensity profile at the specified time step, given the subfield of interest. We derived $\Delta\lambda$ from the difference in spectral position of the line core from the intensity profile at each pixel in the subfields and λ_0 .

The low spectral resolution of the SST data makes it difficult for us to derive the Doppler shifts with the precision needed for the current study. We resolved this issue by interpolating between the spectral points to increase the number of data points. We chose one thousand interpolation points. From there, the intensity line profiles were fitted with two profiles, Gaussian and Voigt, to achieve a smoother representation. The Gaussian profile is described in Equation 3

$$G(x) = A \cdot \frac{1}{\sigma \sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{x - \lambda_c}{\sigma}\right)^2\right] + I_c, \quad (3)$$

where λ_c is the center position of the peak and σ is the standard deviation. A is the amplitude of the spectral line and we included the continuum intensity I_{cont} as the base of the line since we have absorption lines in our data. The Voigt profile is a convolution of a Gaussian and Lorentzian profile, related to the real part of the Faddeeva function $w(z)$,

$$V(x; \sigma, \gamma) = A \cdot \frac{\text{Re}[w(z)]}{\sigma \sqrt{2\pi}} + I_c, \quad \text{where } z = \frac{x + i\gamma}{\sigma \sqrt{2}}. \quad (4)$$

The parameter σ represents the standard deviation of the Gaussian component and γ represents the half-width at half-maximum (HWHM) of the Lorentzian component.

We utilized `curve_fit` from SciPy (Virtanen et al. 2020) to apply a Gaussian and Voigt profile to the spectral lines. The method uses non-linear least squares to fit a function to data and is based on initial guesses as input. The initial guesses are derived from the intensity profiles we want to describe using a Voigt/Gaussian profile. We estimated λ_c as the mean value of the wavelength range,

$$\lambda = 617.34 \text{ nm}, t[0] = \text{b'08:22:21.619'}$$

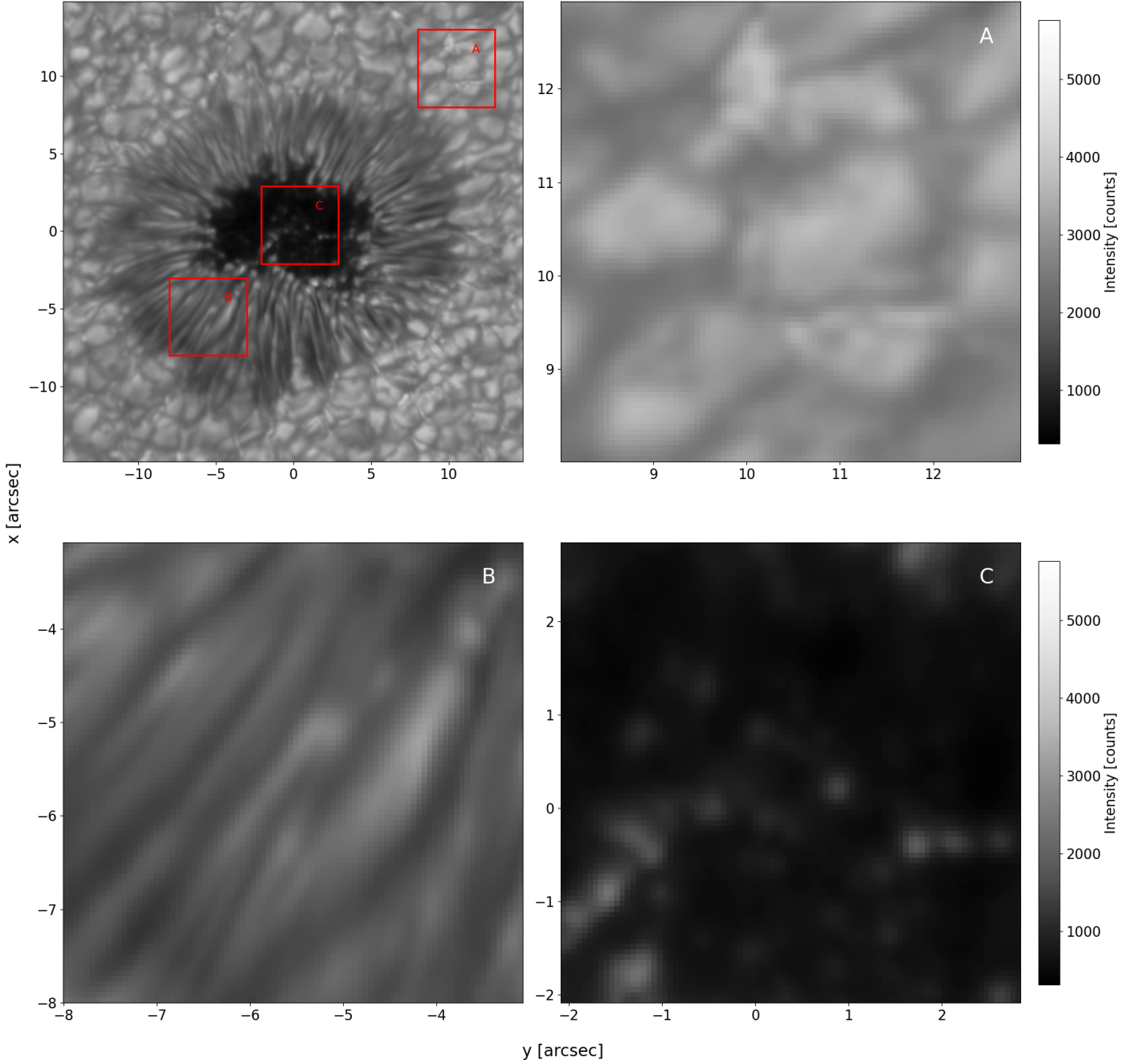


Fig. 2. Intensity map at the initial time step of a sunspot located in AR12770 observed on 2020-Aug-07 (*upper left panel*). Subfield A (*upper right panel*) is a region outside the sunspot with a visible granulation pattern. Subfield B (*right panel*) is located in the sunspot penumbra and subfield C is located in the sunspot umbra.

I_{cont} as the maximum intensity value, A as depth at the minimum intensity value relative to the continuum intensity value and σ as the full width at half maximum (FWHM). In many cases, there were no data points where the full width at half maximum is located on the spectral lines. Rather than increasing the number of interpolation points, we allowed the code to use the closest neighboring points as the FWHM. To define the closest neighboring points, we define the limit of how many intensity counts away from the true FWHM the code can use as the FWHM for fitting a profile. This will be referred to as the FWHM-threshold and is illustrated in [Figure 3](#). The FWHM-threshold is set to the lowest value that enables the code to identify an FWHM in that particular analysis step. We present the different thresholds used for different analysis steps in [Table 2](#).

While the Voigt profile provides a more accurate representation of spectral lines and encapsulates several physical broadening mechanisms, its complexity makes the computation more demanding. We discovered that fitting a Gaussian to a spectral line is ten

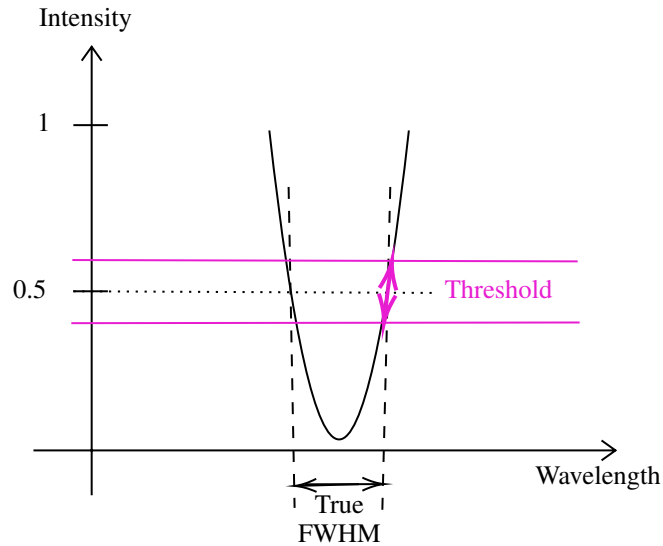


Fig. 3. Illustration of the FWHM-threshold: The black solid line represents an absorption line, while the black dotted line indicates the half-maximum intensity level. The distance between the two vertical black dashed lines corresponds to the true full width at half maximum (FWHM) of the line. The FWHM-threshold is defined as the region bounded by the two horizontal pink lines, highlighting the intensity range used to quantify the FWHM.

Analysis step	Threshold [Intensity counts]
Error analysis	5
Spatially averaged intensity profiles	8
Doppler maps	15
Equivalent width	15

Table 2. FWHM-threshold used for different analysis steps in this work.

times faster than fitting a Voigt profile. Additionally, we find the Gaussian is sufficient for instances where we are only interested in the line core. We resort to fitting a Voigt profile when we are interested in the line wings, particularly when determining the equivalent width (EW).

2.5. Spectral line shape and line broadening

The Gaussian component of a line profile arises from the assumption that the velocities of absorbing or emitting particles follow a Maxwell-Boltzmann distribution, which is valid for particles in thermal equilibrium. This leads to Doppler broadening, a temperature-dependent effect that reflects the thermal motion of atoms or ions. As the temperature increases, particle velocities become more widely distributed, resulting in a broader Gaussian core of the spectral line. The spectral shape of a Lorentzian describes the spread of energy, related to the finite lifetime of an electron in an excited state (Rutten 2015). The finite lifetime comes from the description of natural broadening, where Heitler (1954) demonstrates that if an electron in an excited state spontaneously decays, that state has a finite lifetime. Therefore, the Lorentzian line shape relates to the spontaneous radiation process (Gomez et al. 2022). In practice, the Lorentzian is also a result of collisional damping due to neighboring particles. Thus, the shape of the Voigt profile is determined by broadening mechanisms such as Doppler broadening, natural broadening and collisional damping (Rutten 2015). For our study, the most relevant broadening mechanisms are Zeeman broadening, due to the presence of magnetic fields, and Doppler broadening, arising from thermal and bulk plasma motions. However, Asplund et al. (2000a) emphasizes the importance of non-thermal broadening to accurately capture the full width of the spectral line. This is in terms of synthesized spectral lines and how well they align with observed spectra. Non-thermal broadening is beyond what is expected from thermal motion alone. It is caused by the solar environment, such as turbulence, waves and unresolved flows.

2.6. Equivalent width

Suppression of convective motion due to magnetic fields influences the solar environment, which further impacts how strongly absorption lines manifest. A good measure for the strength of an absorption line is the EW. The EW of a spectral line is defined as

$$EW = \int \frac{I_{\text{cont}} - I_{\lambda}}{I_{\text{cont}}} d\lambda, \quad (5)$$

where I_{cont} is the continuum intensity, I_{λ} is the line intensity at a wavelength λ . It has its name because it is equivalent to the width of a rectangle with height spanning from zero to the continuum intensity, blocking out the same amount of the spectrum (see illustration in Figure 4). We define the continuum intensity, lower and upper integration range of a spectral line in the following

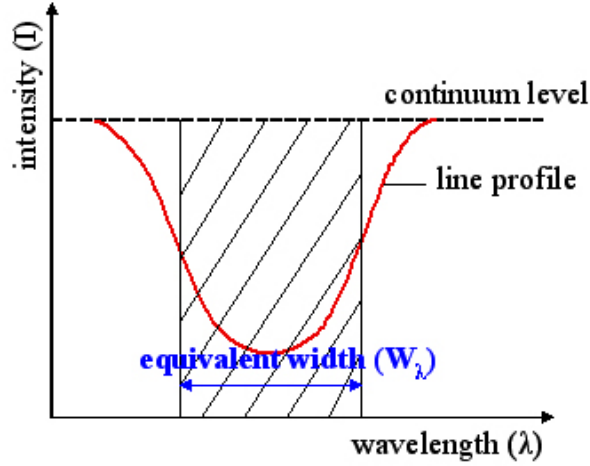


Fig. 4. Diagram of the EW. The red line represents the spectral line profile, while the black dashed line indicates the continuum level. The width of the shaded box with diagonal lines corresponds to the EW. Figure retrieved from Wikipedia (2025).

manner:

- *Continuum intensity.* We define the continuum intensity as the peak intensity value of the histogram at a given time and/or xy-position, using `numpy.histogram`.
- *Line center.* The line center is defined at the spectral position where the minimum intensity of the line is located.
- *Lower limit.* Starting at the blue wing and moving towards the line center, we look for the first intensity value smaller than the continuum intensity value. The first point to fulfill this condition is the lower limit of the integration range.
- *Upper limit.* Starting at the red wing and moving towards the line center, we look for the first intensity value smaller than the continuum intensity value. The first point to fulfill this condition is the upper limit of the integration range. By redefining the continuum intensity to be

To calculate the EW, we use `numpy.trapz` for the integration.

3. Results

3.1. Intensity Structure of Sunspot and Surrounding Regions

In Figure 2, we present the intensity map for the sunspot located in active region AR12770. This figure includes the full field of view (FOV) and highlights three subfields (A, B, and C) captured at the initial time step of the observation sequence. These subfields were selected to represent regions with varying magnetic field strengths. A time-lapse of this map is available on GitHub¹. We begin by examining the intensity structures within each of the three subfields. Subfield A (upper right panel of Figure 2) lies in a magnetically quiet region of the photosphere and displays the classic granulation pattern characteristic of convective processes. These granules appear as bright, irregularly shaped features surrounded by dark intergranular lanes. The granules originate from the

¹ https://github.com/rebkabob/solar_and_stellar_physics/blob/main/figures/intensity_A_B_C/intensity_A_B_C.gif

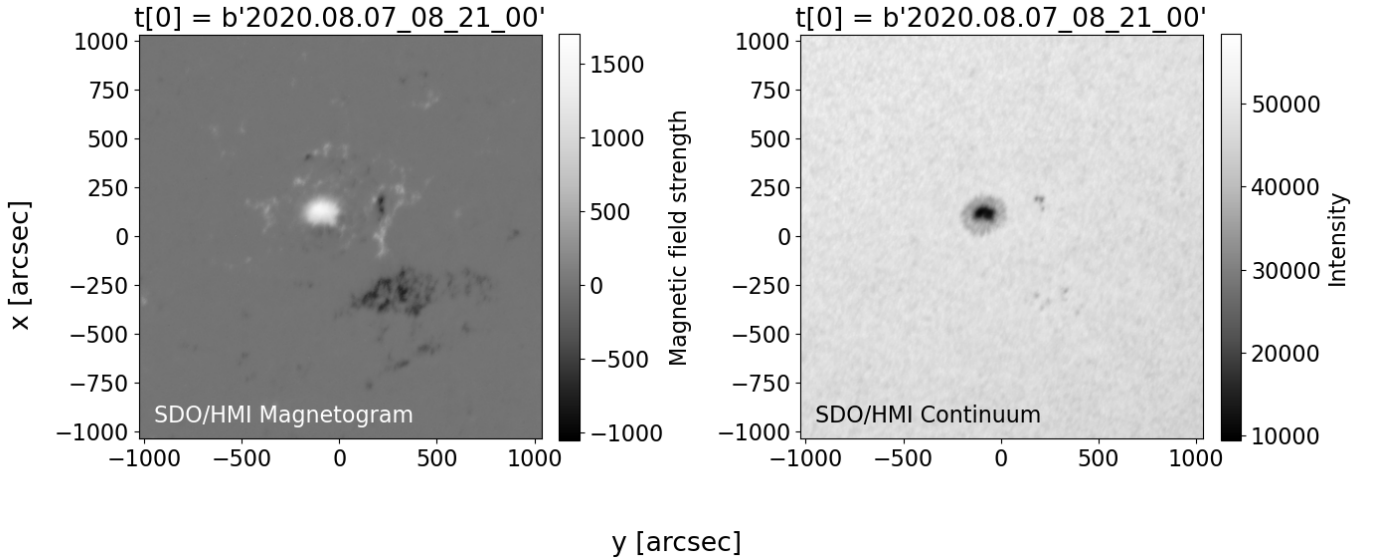


Fig. 5. Observations from the Helioseismic and Magnetic Imager on board the Solar Dynamics Observatory on 2020-08-07. Left: Line-of-sight magnetogram showing magnetic field strength, where white and black indicate opposite polarities. Right: Continuum intensity image showing a sunspot located in AR12770.

rising and sinking plasma and are seen as bright blobs with surrounding dark intergranular lanes. By examining the video, we find the granulation pattern to be highly dynamic with their lifetime in the order of ~ 10 -15 min. In contrast, subfield B (lower left panel of Figure 2) is located closer to the sunspot boundary where the magnetic field becomes stronger. The granulation pattern in this region is less pronounced and the intensity appears more uniform. The observed plasma motion is reduced along the LOS and is likely dominated by horizontal flows, which are less efficiently captured in intensity maps. This subdued appearance suggests that the stronger magnetic field in this area partially suppresses vertical convective motions, altering the standard granulation pattern. Subfield C (lower right panel of Figure 2) is situated within the umbral core of the sunspot, where the magnetic field strength reaches its peak. Here, the intensity is significantly lower, and the region appears dark. Aside from a few isolated bright points, possibly umbral dots or signatures of magnetoconvection, the plasma shows little evidence of motion. This suggests that strong magnetic fields in the umbra effectively suppress convection, consistent with prior observational studies.

In these intensity maps, we see a correlation between intensity and magnetic field strength. Dark regions, such as subfields B and C, exhibit less apparent plasma motion compared to brighter regions like subfield A. This is directly related to the magnetic field strength, seen in Figure 5 from the Helioseismic and Magnetic Imager (HMI) on board the Solar Dynamics Observatory (SDO). From a rough estimate using SDO/HMI magnetograms, we find that strong magnetic fields are present in subfield B (approximately 50–90 G) and subfield C (>1300 G), compared to the much weaker field in subfield A (around ~ 15 G). The consequence of having a strong magnetic field present is that less energy gets transported and released in the photosphere. This, in turn, leads to the dark appearance of the umbra and penumbra, providing evidence that strong magnetic fields suppress how convective motion manifest itself on the photosphere.

3.2. Voigt Profile Versus Gaussian Profile

To better deal with uncertainties in the data and derive properties with high precision, we fit spectral lines with both Voigt and Gaussian profiles. In the left panels of Figure 6, we present the temporally and spatially averaged intensity profiles for subfields A, B and C, along with their corresponding Voigt and Gaussian fitted profiles. To evaluate and measure the extent of their differences, we calculated the relative error of both profiles fitted to the same line and compared the results. The relative errors of the fitted profiles are shown in the right panels of the same figure. By the visual comparison on the left-hand side of the figure, we see that the Voigt profile aligns more closely to the spectral line than the Gaussian profile. That is especially visible in the line wings and line depth. The position of the line core for both profiles is identical. Thus, we confirm that Gaussian components dominate the line core, while the Lorentzian components become more important when modeling the line wings. We find that the relative error is highest near the line wings. As expected, the relative error is also greater for the Gaussian profile compared to the Voigt profile, reflecting

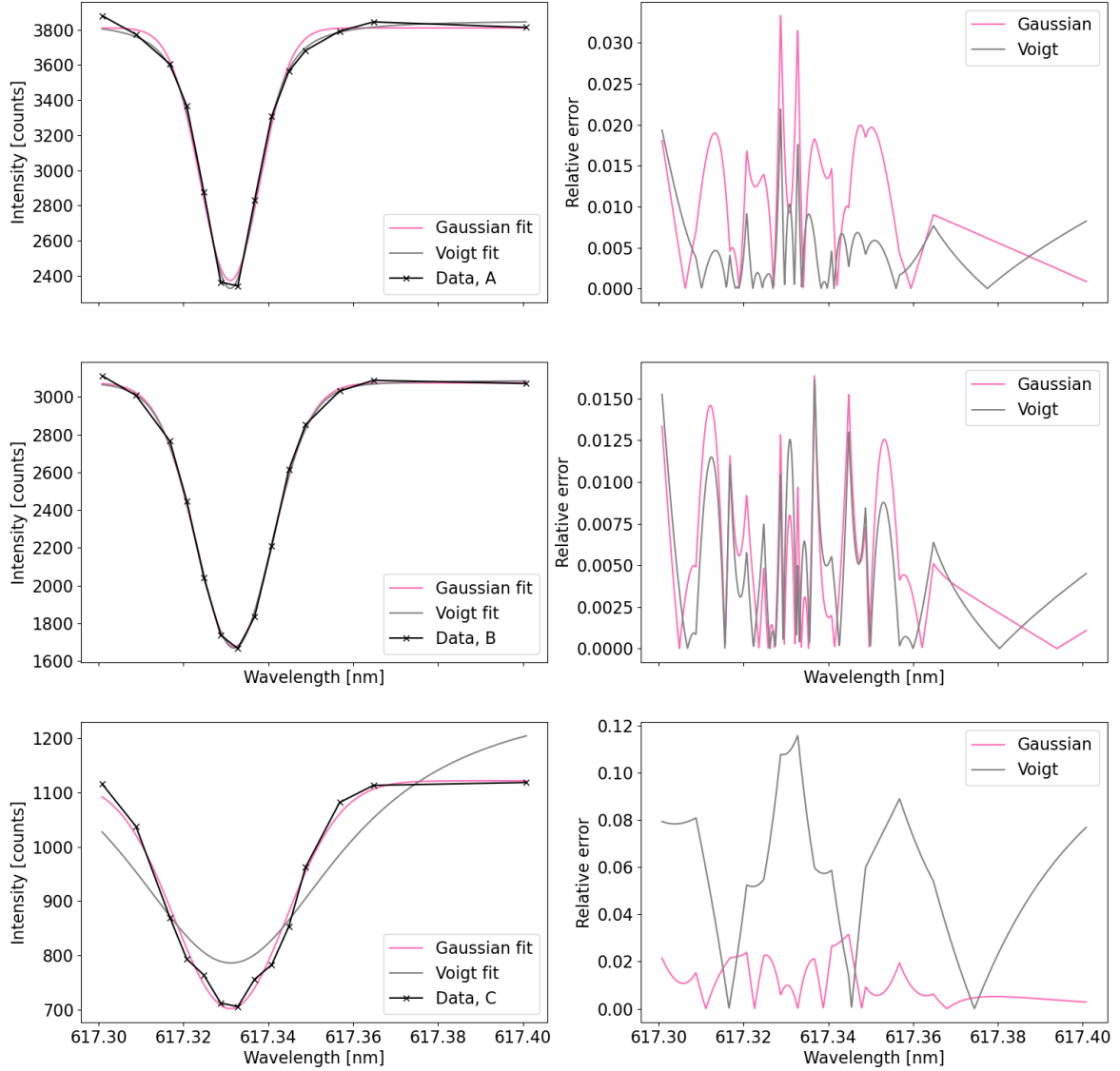


Fig. 6. Data points of temporally and spatially averaged intensity profile (black) for subfield A (*upper left panel*), subfield B (*left-hand side, middle panel*) and subfield C (*lower left panel*) along with the Voigt (gray) and Gaussian (pink) fitted profiles. The relative error for the fitted profiles is shown in the upper, middle and lower right panels for subfields A, B and C, respectively.

the latter's greater complexity and ability to better capture the full line shape. However, the performance of the Voigt and Gaussian profiles is highly sensitive to the chosen FWHM-threshold value. With a threshold of 5 intensity counts used in the error analysis, Voigt profiles provide a better fit for narrow lines (as seen in the upper and middle left panels), while Gaussian profiles perform better for broader lines (lower left panel). When the threshold is increased to 8 intensity counts, the Voigt profile consistently outperforms the Gaussian in all cases. This demonstrates that the threshold value can introduce numerical error into the analysis. Henceforth, where the spatially and temporally averaged intensity profiles are used, the FWHM-threshold is set to 8 intensity counts. All this goes to prove that Voigt profiles are far superior to Gaussian. As already established, the line core of a spectral line is dominated by Gaussian components. The line core is also the most important part of a spectral line when calculating Doppler velocities. Even though the Gaussian profile is less accurate than the Voigt profile, it is sufficient for this instance. One advantage of the Gaussian

Subfield	Voigt [s]	Gaussian [s]
A	0.0237	0.0024
B	0.0324	0.0021
C	0.0292	0.0026

Table 3. Computation time of fitting a Voigt versus a Gaussian profile to the temporally and spatially averaged intensity profile for subfield A, B and C.

profile is that its simplicity also makes for faster computation. We confirm this by timing the Voigt and Gaussian curve fitting methods against the temporally and spatially averaged intensity profiles for subfields A, B and C. The computation times are listed in Table 3. At the cost of lower accuracy, the Gaussian method is indeed over ten times faster than the Voigt method. In conclusion, we use the Gaussian profile to calculate Doppler velocities, as it provides a good description of the line core and is more efficient. For EW calculations, however, we employ the Voigt profile, which offers a more accurate representation of the entire line, including the wings.

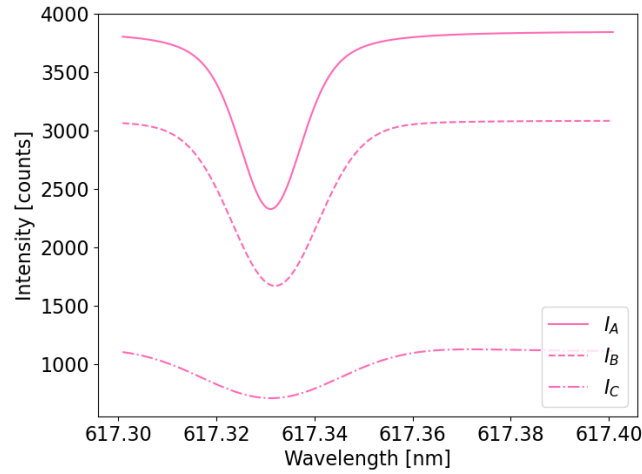


Fig. 7. Comparison between temporally and spatially averaged intensity profile for subfield A (solid line), subfield B (dashed line) and subfield C (dashed dot line).

3.3. Temporally and Spatially Averaged Intensity Profile

Figure 7 shows the temporally and spatially averaged intensity profile for subfields A, B and C. The lines are all fitted with a Voigt profile and FWHM-threshold of 8 intensity counts. There is a noteworthy difference in the continuum levels. The continuum for subfield A is stronger with intensity $I_{c,A} = 3829.96$ counts, followed by the continuum for subfield B at $I_{c,B} = 3069.946$ counts and lastly, subfield C with $I_{c,B} = 1117.58$ counts. This corresponds well to what we observe in the intensity map in Figure 2, where subfield A is the brightest, followed by subfield B and C.

The subtle difference between the lines becomes more prominent when the lines are normalized to 1. This has been done in Figure 8 where each profile is divided by its respective continuum intensity. We find that the spatially and temporally averaged line profile is narrowest in subfield A, deepest in subfield B, and broadest in subfield C. The reason subfield B exhibits the deepest line profile is that the penumbra is a cooler region with less intensity emitted, resulting in fewer atoms being ionized. With more neutral atoms present, absorption increases, leading to a stronger and deeper absorption line. This trend is not observed in the profile of subfield C, suggesting that the strong magnetic field induces significant Zeeman splitting, which counteracts the absorption and broadens the line instead of deepening it. For all three profiles, a trend of increasing line width is observed as the magnetic field strength increases. We quantify this in subsection 3.5. We see a clear systematic blueshift in all three profiles in the figure, suggesting bulk downflows in the atmosphere regions where these cores were formed.

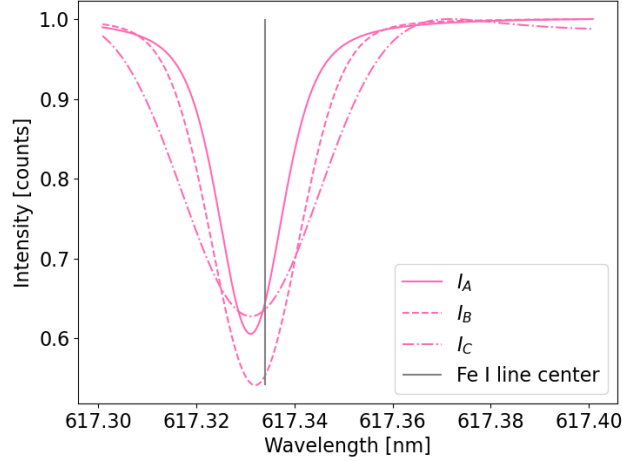


Fig. 8. Temporally and spatially averaged intensity profile for subfield A (solid line), B (dashed line) and C (dashed dotted line), all normalized to 1. The gray solid line shows the line center for the Fe I 617.3 nm line.

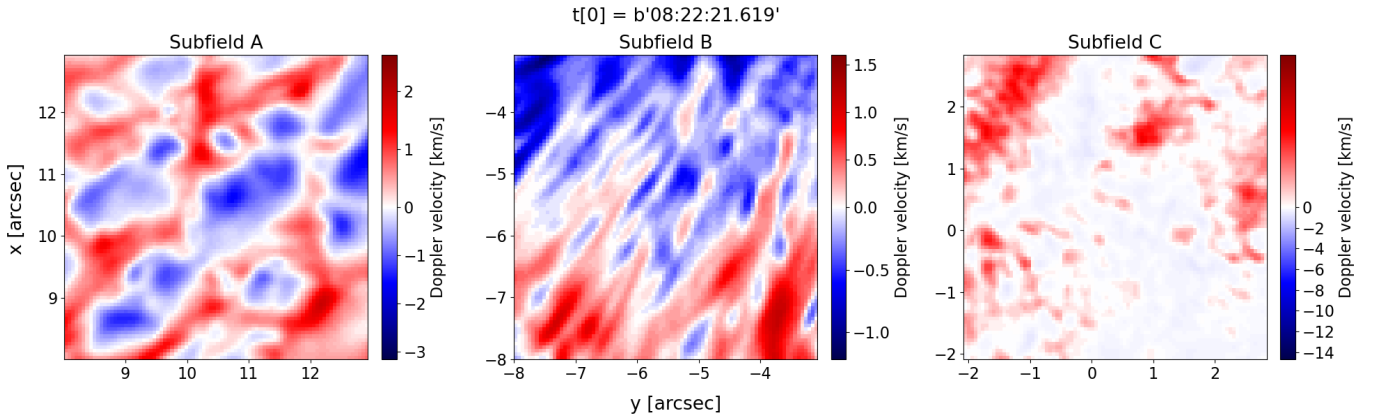


Fig. 9. Doppler map for subfield A (left panel), subfield B (middle panel) and subfield C (right panel) at the initial time step. The blue and red colors in the maps are related to blue- and red-shifted Doppler velocities. White regions are where the Doppler velocities are equal to 0 km/s.

259 3.4. Doppler Velocity

260 We calculated the Doppler velocities for all pixels in the subfields at every time step and created Doppler velocity maps. In Figure 9,
 261 we present the Doppler maps for subfields A, B and C, at the initial time step. The maps were created with a FWHM-threshold
 262 of 15 intensity counts and movies can be found on Github² The blue colors in the map correspond to blue-shifted and negative
 263 velocity values, while the red colors correspond to red-shifted and positive velocities. We observe that subfields A and B exhibit
 264 significantly more up- and downflows than subfield C, which appears to be almost static along the LOS. The shape of the cells in the
 265 left panel of Figure 9 is not so surprisingly the same as the granules seen in the upper right panel of Figure 2. The cells have a dark
 266 blue center, which becomes a lighter blue closer to the edge, meaning the plasma rises more rapidly in the center of the cells and
 267 slows down towards the edges. The red regions surrounding the blue cells are due to red-shifted plasma velocities and correspond
 268 to the intergranular lanes where the plasma sinks. However, compared to Figure 2, the granular cells in the Doppler map appear
 269 smaller and this finding is further discussed in subsection 4.3. An interesting observation in the Doppler map of subfield B is the
 270 clear distinction between areas where the plasma is blue-shifted and those where it is red-shifted. An explanation for this separation
 271 may be due to the penumbral filaments. Some penumbral filaments are vertical and extend way beyond the extent of the sunspot,
 272 while other filaments extend up at a lower inclination angle before pointing back at the surface. The latter explains the Doppler
 273 map for subfield B, where the blue-shifted area is due to vertically inclined filaments, and red-shifted areas are where the filaments
 274 dive back down the surface. Additionally, we know that flows along penumbral filaments are referred to as Evershed flow, from the
 275 work of Evershed (1909). Evershed flow is seen as radial movement outwards from the sunspot center, essentially horizontal plasma
 276 motion. In Doppler maps, the sunspot is blue-shifted on the center side and red-shifted on the limb side. We confirm this effect can

² https://github.com/rebkabob/solar_and_stellar_physics/blob/main/figures/doppler_map/doppler_map.gif

277 be observed for our sunspot in the following [video](#)³.

278 The correlation between the intensity map and the Doppler maps varies with subfields. For subfield A, we see that high-intensity areas
 279 correspond to blue-shifted Doppler velocities, areas of lower intensity are associated with red-shifted velocities. In high-intensity
 280 areas, the plasma rises rapidly, resulting in high blue-shifted velocities, and releases energy at the surface, which in turn leads to
 281 higher intensity. As for subfields B and C, the correlation is more difficult to determine due to the low intensity contrast in the
 282 penumbra and umbra. In those cases, the Doppler maps prove more useful for studying plasma dynamics, as they provide clearer
 283 insight into LOS motions despite the reduced intensity contrast. To understand how the suppression of convective motion relates to
 the measured Doppler velocity, we also visualize the Doppler velocities as histograms in [Figure 10](#).

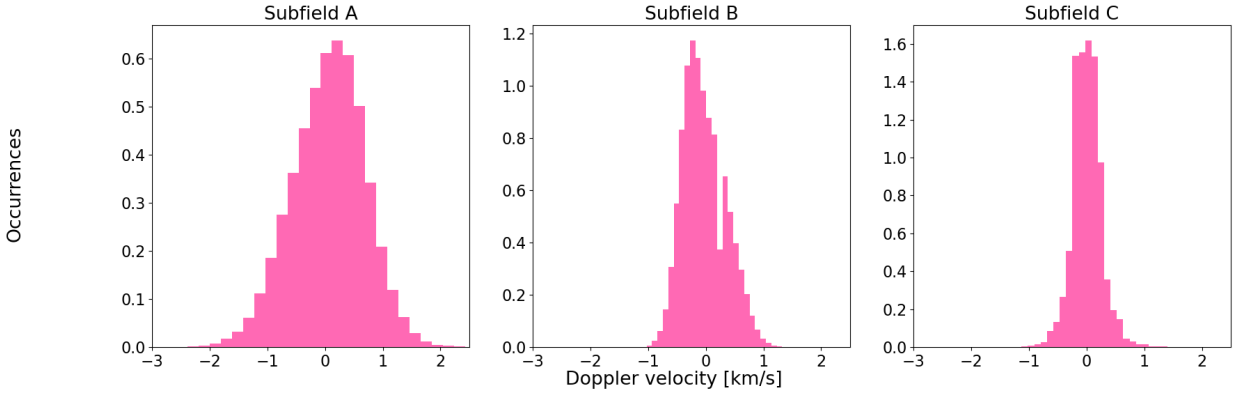


Fig. 10. Histogram of measured Doppler velocities for every pixel at every time step within subfield A (*left panel*), subfield B (*middle panel*) and subfield C (*right panel*). The distributions are normalized so that the area under the histograms sums to one and the number of bins used is 30 for subfields A and B, 150 for subfield C.

284

Subfield	Min [km/s]	Max [km/s]	Mean [km/s]	Standard deviation [km/s]	RMS [km/s]
A	-3.159	2.624	0.067	0.635	0.639
B	-1.215	1.604	-0.027	0.365	0.365
C	-14.677	1.847	0.004	0.256	0.256

Table 4. Statistical properties of Doppler velocity distributions in three selected subfields (A, B and C) at the initial time step. Values represent the minimum and maximum velocities, as well as the mean, standard deviation, and root-mean-square (RMS) of the Doppler velocity distribution in each region.

285 The minimum and maximum value, along with root mean square (rms) and mean value of the distributions in [Figure 10](#), are
 286 shown in [Table 3.4](#). Subfield A is the most dynamic subfield with Doppler velocities ranging from -3.159 km/s to 2.624 km/s.
 287 Followed by subfield B with Doppler velocities ranging -1.215 km/s to 1.604 km/s. For subfield C, the lowest measured Doppler
 288 velocity was -14.677 km/s, but did not appear in the histogram. Upon further inspection, it was identified as an outlier. After filtering
 289 outliers, we find the Doppler velocity range for subfield C to be from -2.429 km/s to 1.847 km/s. The presence of high velocity
 290 dispersion in subfield A, almost up to 3 km/s, suggests that the plasma motion in this subfield is predominantly along the LOS,
 291 either as an upflow or downflow. The somewhat symmetric distribution suggests the observations cover the lifetime of the granules,
 292 meaning the time during which the granules appear due to rising plasma and dissipate as the plasma sinks into the intergranular
 293 lanes. In contrast, subfield B displays a narrower distribution with a peak at -0.15 km/s. The tall and narrow distribution suggests
 294 less plasma movement along LOS. As previously established, the penumbral filaments are predominantly horizontal. Thus, we
 295 would not observe significant variations in Doppler velocity with our viewing angle of $\mu = 0.87$. This would result in a narrow
 296 distribution of Doppler velocities for subfield B, which is consistent with our observations. In subfield C, we find the distribution to
 297 be even narrower than in subfield B, with an average measured Doppler velocity of 0.004 km/s. Although the distribution is more
 298 concentrated, the overall velocity range remains somewhat similar to subfield B.

299

³ Courtesy of Luc Rouppe van der Voort.

3.5. Equivalent Width

We measured the EWs of the spatially averaged intensity profile for the subfields at every time step. The temporal averaged EW of subfield A is measured to be 68.563 mÅ, while subfield B exhibits a broader profile with a temporal averaged EW of 104.05 mÅ and subfield C has the broadest profile of 114.52 mÅ. We were also curious to explore whether there is a relation between EWs and Doppler velocities. In Figure 11 we present scatter plots of EW versus Doppler velocity for all subfields at the initial time step.

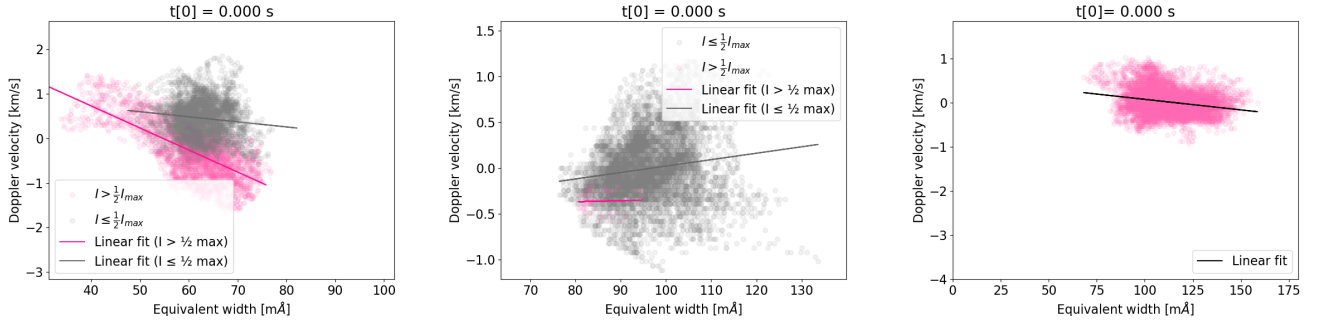


Fig. 11. EW versus Doppler velocity measured for every pixel at the initial time step. Subfield A is shown in the left panel, subfield B in the middle panel and subfield C in the right panel. For subfields A and B, the scatter points are color-coded: pink represents pixels with intensity values above $I > \frac{1}{2}I_{\max}$, while gray represents those below. Linear regression lines are overlaid in matching colors to indicate trends within each intensity group. No color coding has been applied to subfield C.

Each marker corresponds to a pixel in the subfield at the given time step and a low-opacity marker was chosen, making it easier to identify the high-density areas of the figure. For subfield A and B, the scatter points are color-coded: pink for pixels with intensity values above $\frac{1}{2}I_{\max}$ and gray for pixels below. In the case of the granulation pattern, the pink scatters relate to granule pixels and the gray relates to intergranular pixels. For subfield C, the color-coding did not reveal any significant additional information and was therefore omitted. We performed a linear regression on the data points at each time step to identify any particular trends. Full movies showing the temporal evolution of EW versus Doppler velocity, as well as the linear regression, can be found on [GitHub](#). We note that due to the scattering of the data points, the reliability and meaning of linear regression become limited. By color-coding the pixels, we are able to detect a clear trend for the pixel groups in subfield A, but not so much for the other subfields. In subfield A for the granule pixels, we detect a strong negative correlation where stronger lines accompany the strengthening of convective blue shifts. For the intergranular pixels, the linear regression shows a much weaker negative correlation between EW and Doppler velocities. We also find that the intergranular pixels are mostly grouped near large EWs, indicating that downflow is associated with stronger lines. For subfield B, we find that the linear regression indicates no clear correlation. There are significantly more scattered points for this subfield than the previous one, making it difficult to draw a conclusion. Lastly, for subfield C, not only is there a weak correlation, but the slope varies rapidly through time. Again, making this very hard to conclude. The results could be improved by implementing a threshold to filter out the outliers. Additionally, applying several color-codings could help clarify the observed scatter by revealing underlying patterns or correlations within different intensity or velocity regimes.

4. Discussion

4.1. Alteration of spectral shape

We measured the EW to determine the spatial and temporal line strength. For subfield A we measured $W_\lambda = 68.563$ mÅ, which is in good agreement with [Allende Prieto et al. \(2002\)](#) who measured $W_\lambda = 69$ mÅ and [Asplund et al. \(2000b\)](#) which obtained $W_\lambda = 6.74$ pm. Both were determined from the same spectral line as in our work, but from a 3D model. The minor deviations are caused by the choice of integration range and the fact that the EW was measured from observational spectra. However, the agreement between our results and those from the literature demonstrates the viability of our method. For subfield B and subfield C, we measured $W_\lambda = 104.05$ mÅ and $W_\lambda = 114.52$ mÅ, respectively, suggesting that a strong magnetic field alters the shape of a spectral line. In comparative study by [Criscuoli et al. \(2013\)](#) of Fe I 630.2 nm and Fe I 617.3 nm line, the authors found evidence that the dependence of the magnetic field of the EW is negligible at the investigated magnetic flux values ranging between 0-1000 G. The variation of average line shape occurs due to temperature variations induced by the presence of magnetic field concentrations. However, they found that the magnetic flux increases the FWHM, where larger variations were found for Fe I at 630.2 nm. Therefore, the Fe I 630.2 nm line is more favorable for investigating magnetic field variation, while the Fe I

617.3 nm line is more suitable for investigating the effects of temperature variations. This finding is supported by Fabbian et al. (2010) whose paper is based on simulations from the Copenhagen stagger code (Nordlund & Galsgaard 1995, Carlsson et al. 2004, Stein & Nordlund 2006, De Pontieu et al. 2006) to explore the effect of the magnetic field on determining solar element abundances. Their results show that changes in the EW of visible Fe I lines have a small contribution from the Zeeman effect. The dominant contribution arises from indirect magnetic effects, specifically changes in stratification leading to different amounts of absorption in the line core.

4.2. Broadening effects

As shown in section 3, the temporally and spatially averaged intensity profile for subfield B and C displays a broader shape compared to that of subfield A. Since the temperature in the penumbra and umbra is lower than in the quiet Sun, reduced turbulence is expected, leading to less Doppler broadening in the spectral line. As a consequence, the observed line broadening is more likely attributed to the Zeeman effect resulting from the strong magnetic field. The Fe I 617.3 nm line is a magnetically sensitive line with an effective Landé factor of 2.50. In the case of high magnetic field strengths, the line is split so strongly that its Zeeman σ components begin to overlap mutually (Lozitsky et al. 2020). The authors state that this is especially true in the sunspot umbra, where numerous and variable blends can occur. Magnetic fields in the range of 7-8 kG are close to the upper limit of detectable fields when using spectral lines with large Lande factors. To measure even stronger magnetic fields they suggest using spectral lines with small Lande factors ($g_{\text{eff}} \leq 1$). As previously discussed, both Criscuoli et al. (2013) and Fabbian et al. (2010) demonstrate that variations in EW are more strongly influenced by temperature than by magnetic field strength. In contrast, Tritschler et al. (2004) emphasizes that FWHM is primarily affected by changes in the magnetic field, suggesting it may serve as a more suitable parameter than EW for future investigations.

4.3. Shapes in the Doppler maps

The discrepancy in the appearance of granules in intensity and Doppler velocity maps arises from the different physical processes each map represents. Intensity maps reflect the distribution of temperature and opacity within the FOV, highlighting convective structures. On the other hand, Doppler velocity provides information on plasma motion along the LOS. Given that the Doppler map only provides information along the LOS, we lose information for other inclination angles. The loss of information is most likely evident as a difference in granule size when comparing the two maps. Nesis et al. (2002) demonstrated that the intensity map and Doppler velocity map provide different granular dynamical times. The former gives the typical duration of enhanced intensity at a position, while the decay of Doppler velocity defines the latter. The authors state both are needed to describe the granular evolution, but address two partially different processes associated with the intensity and velocity.

4.4. Doppler velocity

Ref.	Doppler velocity [km/s]	Solar disk position
(-)	[-3.15, 2.62]	Subfield A
(1)	[-3.3, 2.0]	(31, 200), (0, 200), (0, -200)
(2)	[-2.16, 3.0]	7 regions near disk center

Table 5. Doppler velocities measured in quiet Sun from our analysis, presented alongside values reported in the literature. Table adapted from Table 1 of Falco et al. (2017). Ref.: Our work, (1) Jin et al. (2009), (2) Yu et al. (2011).

Table 5 presents our measured range of Doppler velocities in subfield A, compared with values previously reported in the literature. In our data, Doppler velocities in this weak-field region range from -3.15 km/s to 2.61 km/s. These velocities are consistent with those expected in quiet Sun granulation, where convective upflows and downflows dominate the velocity field. Interestingly, we observe an asymmetry in the distribution of Doppler velocities: the slope of the distribution toward positive (redshifted) velocities is steeper than that toward negative (blueshifted) velocities. This contrasts with the findings of Hirzberger (2002) and the authors report velocities of the granules in the quiet sun are almost symmetrical around 0 km/s, indicating that the upflow and downflow areas are almost equal. Despite the slope asymmetry in our case, the overall distribution in subfield A remains centered around 0

km/s, which supports the broader conclusion that the granulation pattern retains its fundamental convective character in the absence of strong magnetic fields.

Jin et al. (2009) showed that the granular motion is within the range of -3.3 km/s and 2.0 km/s. While Yu et al. (2011) reported that Doppler velocities in quiet Sun granules range from -2.16 km/s to 2.98 km/s. Overall, our results are in good agreement with previously reported values in the literature. There are minor deviations, which may be influenced by the analysis method, particularly by the choice of FWHM-threshold. For future work, the analysis could benefit from other means of measuring Doppler velocity than the one presented here. One alternative would be to determine the position of the line core from a parabolic fit around the minimum intensity value. For instance, Tritschler et al. (2004) did the fit of 15 points centered around the minimum of the observed profile. Another alternative would be to measure Doppler velocities from line bisectors.

Ref.	Doppler velocity [km/s]	Heliocentric angle [°]
(-)	[-1.21, 1.60]	29.54
(1)	[-1.3, 1.8]	23

Table 6. Measured velocity of Evershed flow from our analysis, presented alongside values reported in literature. References: (-) Our work, (1) Tritschler et al. (2004).

The flow field in the sunspot penumbra leads to line shift and line asymmetry. The displacement is due to radial movement outward from the sunspot center, essentially horizontal plasma motion. As discussed in section 3, the plasma flow in the penumbra is predominantly horizontal, a phenomenon known as the Evershed flow. For subfield B located in the penumbra, we measured the Doppler velocity range between -1.21 km/s (blue-shift) and 1.60 km/s (red-shift). Tritschler et al. (2004) reported maximum red-shift value reached up to 1.8 km/s and the largest blue-shift value reached absolute values up to 1.3 km/s for a sunspot located at 23° off the disk center. For comparison, our target sunspot is located at 29.54° from the disk center. We note that Doppler shifts also increase with heliocentric angle if there is a strong horizontal component (Johannesson 1993). Although the measured velocities for subfield B are lower than for subfield A, we cannot point to suppression of convection as the cause. This is because the Evershed flow is primarily horizontal, making it difficult to study using LOS measurements at low heliocentric angles.

We would expect the convective blueshifts to be strongly reduced in the penumbra based on literature (e.g. Löhner-Böttcher et al.

Ref.	Convective blueshift [m/s]	Average Doppler shift [m/s]
(-)	-2429	4
(1)	[-61.8, -15.7]	-29.8

Table 7. Doppler velocities measured in the sunspot umbra. Ref.: (-) Our work, (1) Löhner-Böttcher et al. (2018).

2018, Schüssler & Vögler 2006). In Table 7 we find the measured convective blueshift and average Doppler shift in the sunspot umbra from our work and from the work of Löhner-Böttcher et al. (2018). To clarify the amount of convective motion in the darkest part of umbrae, Löhner-Böttcher et al. (2018) measures Doppler velocities for four different sunspots (namely NOAA 12150, 12149, 12146 and 12109). The authors find a significant reduction in convective blueshifts, whilst we do not. Given the similar velocity range for subfields B and C, we suspect the size of subfield C includes parts of the penumbra in addition to the umbra. This overlap could influence the Doppler velocity distribution. For improved accuracy, future analyses should consider using a smaller subfield C that more clearly isolates the umbral region.

399
400

4.5. Correlation between EW and Doppler velocities

We examined the correlation between the EW and Doppler velocity to investigate its implications for convective motion. Our analysis revealed that stronger lines in subfield A accompany the strengthening of convective blue shifts, seen as a negative slope in the left panel Figure 11. This holds for the granular pixels, which are color-coded in pink. For the intergranular pixels, the correlation is also negative but relatively weak. However, this trend is inconsistent with findings reported in the literature. Allende Prieto et al. (2002) found that the strengthening of convective blueshifts is associated with weaker lines (see their Figure 4). Since weaker lines probe deeper layers of the solar atmosphere, this behavior is expected. Granulation with the highest contrast occurs near the

continuum, around $\tau \approx 1$ and the observed continuum originates near the tops of granules. Therefore, we anticipate a stronger convective blueshift for weaker lines that form closer to the continuum-forming layer. The findings of Allende Prieto et al. (2002) are supported by Reinert et al. (2016), who re-investigated convective blueshift in solar absorption lines as a function of line depth and equivalent width. Allende Prieto et al. (2002) report very small intrinsic scatter, significantly smaller than the measurement error, for the Sun. Our data for subfield B do not show a well-defined trend and the scattering in the equivalent width versus Doppler velocity relation may obscure any underlying physical correlation. As for subfield C, we encounter a different issue where the linear regression fluctuates significantly over time. Thus, it prevents us from detecting a clear or persistent trend. We identify that discrepancies may arise from several limitations in our analysis method.

One major limitation involves the line profile fitting procedure. In our study, we used Gaussian profiles to fit the observed spectral lines in order to extract Doppler velocities and equivalent widths. While this approach is computationally straightforward, it does not capture the line wings in a precise manner and affects the measured equivalent width. We recognize that a Voigt profile was superior, but for this instance, it was not chosen due to unresolved numerical issues with the Voigt fitting routine. This requires future investigation.

Another issue lies in the lack of filtering applied to our dataset. We did not implement any thresholding or outlier rejection, which likely introduced additional noise into our results. Based on our results, this would be an important implementation for magnetically complex areas like subfields B and C. Future analysis would benefit from more rigorous filtering and from incorporating additional diagnostics that probe different aspects of line formation and atmospheric structure.

A further complication arises from the inherent three-dimensional structure of the solar atmosphere. We are dealing with 3D stratified plasma, which becomes increasingly opaque. This makes it difficult to determine if the data on the intensity and Doppler velocity provide information at the same layer. Even though both parameters are fundamental to describe convective motion, they translate differently into observables. Our results could be re-evaluated by focusing on the surface where the optical depth $\tau = 1$. Repeating the analysis at this depth would provide a more direct comparison of physical conditions across different regions and may offer clearer insights into the relationship between equivalent width and Doppler velocity.

A more suitable alternative to EW is the use of line bisectors, which offer insight into the vertical velocity gradient within the solar atmosphere. Line bisectors allow us to extract detailed information about convective motion and Doppler velocity. As pointed out by Dravins et al. (1981), each bisector must be terminated at some high intensity level closest to the continuum. That is to avoid possible distant blended lines significantly disturbing the bisector.

Although our results contradict existing literature, they may point toward new findings. To verify this possibility, the same analysis should be performed on a larger sample of sunspots to gather statistically meaningful results.

5. Conclusion

We present an analysis of high-resolution spectroscopic observations of the sunspot in active region AR12770, acquired with the Swedish 1-m Solar Telescope on La Palma. The observations were conducted at a viewing angle of $\mu = 0.87$ using the magnetically sensitive neutral iron absorption line at 617.3 nm. Our study focuses on the intensity structure, spectral line properties, and Doppler velocities across three subfields representing different magnetic environments: subfield A (quiet Sun), subfield B (penumbra), and subfield C (umbra). We find that spectral lines in subfields B and C exhibit significantly broader profiles compared to those in subfield A. This line broadening is most likely attributed to Zeeman splitting due to the presence of strong magnetic fields in the penumbra and umbra, rather than thermal or Doppler broadening, as the local temperatures in these regions are lower than in the quiet Sun.

Doppler velocities measured in the quiet Sun granulation are stronger than those in the penumbra and umbra, which can be explained by the dominance of LOS plasma flows in the granules versus the primarily horizontal flows in the penumbral region. Subfield A shows Doppler velocities ranging from -3.15 km/s to 2.62 km/s, while flow observed in subfield B ranges from -1.21 km/s to 1.60 km/s. Both measurements are consistent with values reported in the literature. For subfield C, we measure flow velocity ranging from -2.429 km/s to 1.847 km/s. Although the overall range is comparable to that of subfield B, the velocity distribution in subfield C is narrower, with a standard deviation that is 0.109 km/s smaller than that of subfield B. Future work includes exploring bisectors as a method to derive Doppler velocities.

The relationship between magnetic field strength and EW remains complex. It is unclear whether the stronger fields in the penumbra are directly responsible for the enhanced EW observed there relative to the quiet Sun. Based on previous studies, future work includes exploring other diagnostics, such as the FWHM, to assess magnetic field effects. Additionally, a more targeted investigation into regions with magnetic field strengths above 1 kG could provide further clarity. To verify the generality of the trends observed

here, a broader statistical analysis involving multiple sunspots would be essential. The use of Gaussian profiles to fit spectral lines introduces limitations, particularly in representing the line wings, where Lorentzian components become significant. Additionally, the FWHM-threshold in our code may introduce numerical error and the fitting method would benefit from creating an automated method for threshold value selection. Overall, this study provides detailed insight into the spectral and dynamic behavior of both quiet-Sun granulation and magnetized sunspot regions. Future improvements in spectral line fitting techniques and Doppler velocity diagnostics will enhance our ability to probe the interplay between magnetic fields and convective dynamics in the solar atmosphere.

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